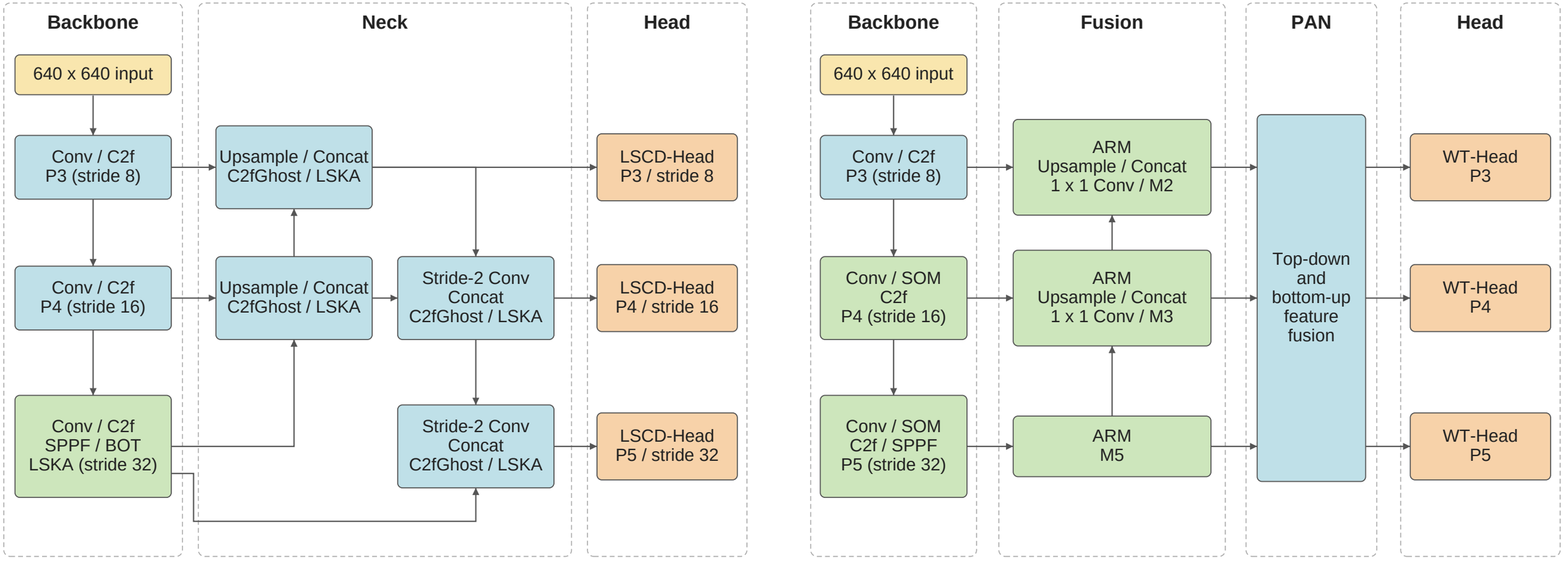
(a) YOLOv8-PD reimplementation
Backbone
Neck
Head
640 x 640 input
Conv / C2f
P3 (stride 8)
Conv / C2f
P4 (stride 16)
Conv / C2f
SPPF / BOT
LSKA (stride 32)
Upsample / Concat
C2fGhost / LSKA
Upsample / Concat
C2fGhost / LSKA
Stride-2 Conv
Concat
C2fGhost / LSKA
Stride-2 Conv
Concat
C2fGhost / LSKA
LSCD-Head
P3 / stride 8
LSCD-Head
P4 / stride 16
LSCD-Head
P5 / stride 32
LSCD convolutions are shared across the three scales.
(b) YOLO-RD reimplementation
Backbone
Fusion
PAN
Head
640 x 640 input
Conv / C2f
P3 (stride 8)
Conv / SOM
C2f
P4 (stride 16)
Conv / SOM
C2f / SPPF
P5 (stride 32)
ARM
Upsample / Concat
1 x 1 Conv / M2
ARM
Upsample / Concat
1 x 1 Conv / M3
ARM
M5
Top-down
and
bottom-up
feature
fusion
WT-Head
P3
WT-Head
P4
WT-Head
P5
Fusion uses Upsample / Concat / 1 x 1 Conv in this study.